WESTERN MICHIGAN UNIVERSITY

College of Engineering and Applied Sciences

Course Name: Lab. Intro. To Microprocessors

Course Number: ECE 2510

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Lab #9

Student:

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Task 1

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Target_1

task3.s | task1.c | task2.s

```c
#include <stm32l476xx.h>

void sum(int *arr1, int *arr2, int *arr3, int size);

int main() {
  int size = 5;                    // size of array
  int arr1[5] = {1, 2, 3, 4, 5};   // array 1
  int arr2[5] = {1, 2, 3, 4, 5};   // array 2
  int arr3[5];                     // result array

  sum(arr1, arr2, arr3, size);     // call sum subroutine

  return 0;
}

void sum(int *arr1, int *arr2, int *arr3, int size) {
  for (int i = 0; i < size ; i++) { // for i = 0 to size do
    int sum = arr1[i] + arr2[i];     // sum
    if (sum < 60) {                  // if sum is less than 60
      sum *= 5;                      // multiply by 5
    }
    arr3[i] = sum;                   // store the sum in result array
  }
}
```

Task 2

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Target_1

Project

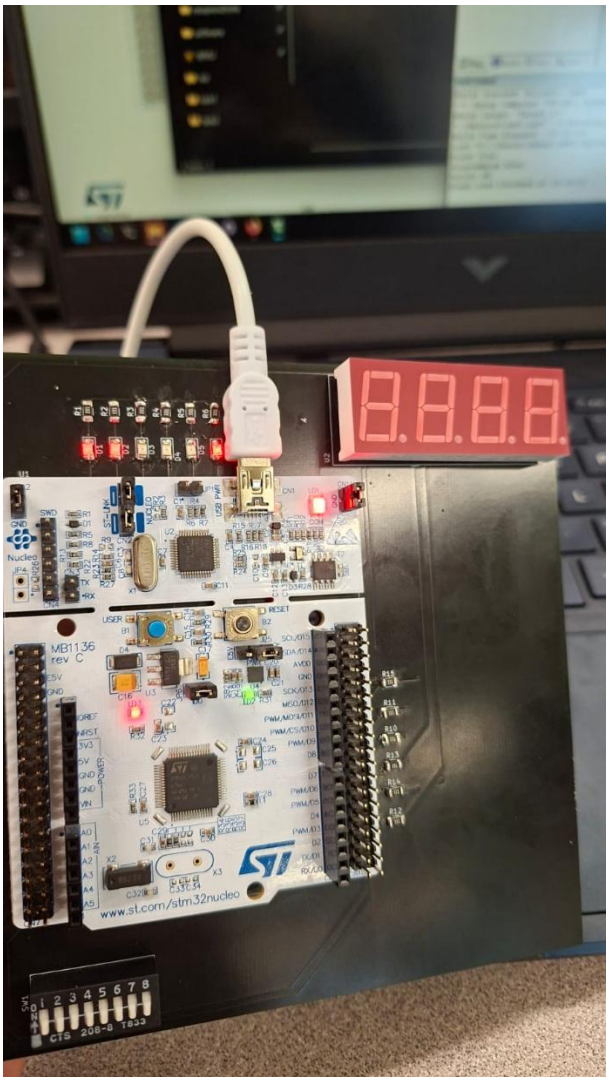
- Project: lab9
  - Target_1
    - Source Group 1
      - task1.c
      - task2.s
      - task3.s
    - CMSIS
    - Device

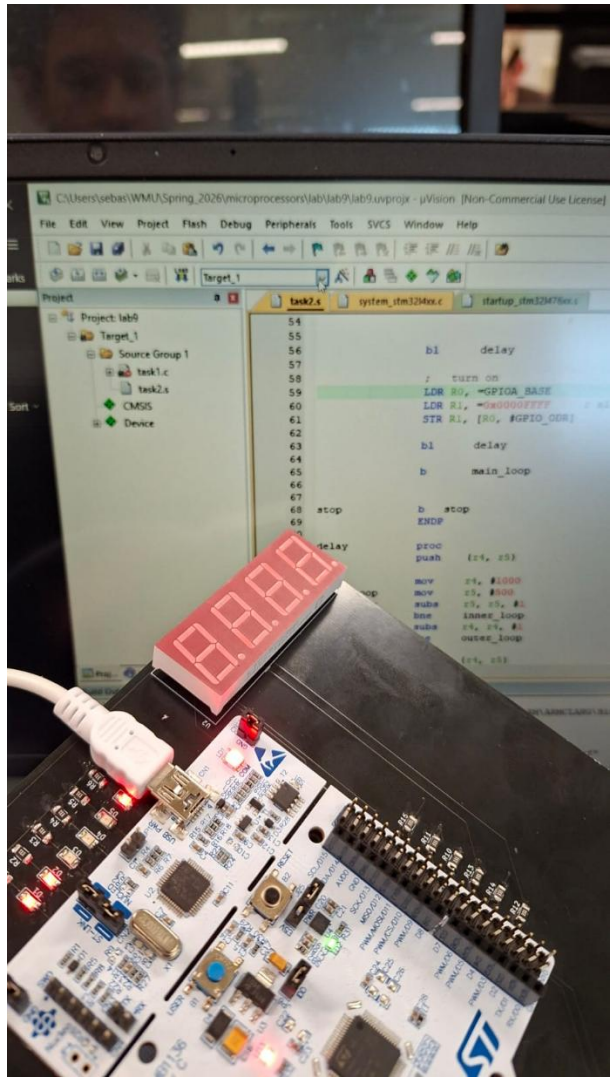
task3.s | task1.c | task2.s

```
                AREA    myData, DATA, READONLY
GPIOA_BASE      EQU     0x48000000
RCC_BASE        EQU     0x40021000

GPIO_MODER      EQU     0x00
GPIO_OTYPER     EQU     0x04
GPIO_OSPEEDR    EQU     0x08
GPIO_PUPDR      EQU     0x0C
GPIO_IDR        EQU     0x10
GPIO_ODR        EQU     0x14

RCC_AHB2ENR     EQU     0x4C

                AREA    main, CODE, READONLY
                EXPORT __main
                ENTRY

PIN             EQU     5
GPIO_BASE       EQU     GPIOA_BASE

Enable_Clock    PROC

                ldr     r2, =RCC_BASE
                ldr     r1, [r2, #RCC_AHB2ENR]
                orr     r1, r1, #1
                str     r1, [r2, #RCC_AHB2ENR]

                bx      lr
                ENDP

__main          PROC
                bl      Enable_Clock

                ldr     r3, =GPIO_BASE
                ;   set pins as output
                ldr     r1, [r3, #GPIO_MODER]
                ldr     r4, =0x0000FFFF
                bic     r1, r1, r4
                ldr     r4, =0x00005555
                orr     r1, r1, r4
                str     r1, [r3, #GPIO_MODER]

                ; test
```

task3.s | task1.c | task2.s

```
                ; test

                ldr     r0, =GPIO_BASE
                ldr     r1, =0x000000FF
                str     r1, [r0, #GPIO_ODR]

main_loop
                ;   turn off
                LDR R0, =GPIOA_BASE
                LDR R1, =0x00000000     ; all 8 LEDs
                STR R1, [R0, #GPIO_ODR]


                bl      delay

                ;   turn on
                LDR R0, =GPIOA_BASE
                LDR R1, =0x0000FFFF     ; all 8 LEDs
                STR R1, [R0, #GPIO_ODR]

                bl      delay

                b       main_loop


stop            b   stop
                ENDP

delay           proc
                push    {r4, r5}

                mov     r4, #1000
outer_loop      mov     r5, #500
inner_loop      subs    r5, r5, #1
                bne     inner_loop
                subs    r4, r4, #1
                bne     outer_loop

                pop     {r4, r5}
                bx      lr
                endp

                END
```





Proof of Task 2 running on board on lab

Task 3

Part B

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Target_1

Project

- Project: lab9
  - Target_1
    - Source Group 1
      - task1.c
      - task2.s
      - task3.s
    - CMSIS
    - Device

task3.s

```
    AREA mydata, DATA, READONLY

; Segment patterns for 0-9, active HIGH common cathode
; Bit order: dp g f e d c b a
seg_patterns    DCB     0x3F    ; 0 = a,b,c,d,e,f
                DCB     0x06    ; 1 = b,c
                DCB     0x5B    ; 2 = a,b,d,e,g
                DCB     0x4F    ; 3 = a,b,c,d,g
                DCB     0x66    ; 4 = b,c,f,g
                DCB     0x6D    ; 5 = a,c,d,f,g
                DCB     0x7D    ; 6 = a,c,d,e,f,g
                DCB     0x07    ; 7 = a,b,c
                DCB     0x7F    ; 8 = a,b,c,d,e,f,g
                DCB     0x6F    ; 9 = a,b,c,d,f,g

; Active LOW digit select - only using digit 1 (CA1)
; CA1=PB8 LOW, rest HIGH
digit_select    DCW     0x0E00

; Register definitions
RCC_AHB2ENR     EQU     0x4002104C      ; enable port clock register
GPIOBEN         EQU     (1<<1)          ; port B 0x00000002

GPIOB_BASE      EQU     0x48000400
GPIO_MODER      EQU     0x00            ; MODER offset
GPIO_ODR        EQU     0x14            ; ODR   offset

;
; adjust to 80000000 if running at 80MHz
ONE_SECOND      EQU     1000
MUX_DELAY       EQU     2000

    AREA mycode, CODE, READONLY
    EXPORT __main
    ENTRY

__main

; --- Enable GPIOA and GPIOB clocks ---
    LDR     R0, =RCC_AHB2ENR
    LDR     R1, [R0]                ; 32 bit value in AHB2ENR
    ORR     R1, R1, #GPIOBEN        ; or with 0x00000002
    STR     R1, [R0]                ; store in AHB2ENR
```

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Target_1

task3.s*

```
__main

; --- Enable GPIOA and GPIOB clocks ---
    LDR     R0, =RCC_AHB2ENR
    LDR     R1, [R0]                ; 32 bit value in AHB2ENR
    ORR     R1, R1, #GPIOBEN        ; or with 0x00000002
    STR     R1, [R0]                ; store in AHB2ENR


; --- Configure PB0-PB11 as outputs (7-seg) ---
    LDR     R0, =GPIOB_BASE
    LDR     R1, [R0, #GPIO_MODER]
    LDR     R2, =0x00FFFFFF
    BIC     R1, R1, R2
    LDR     R2, =0x00555555
    ORR     R1, R1, R2
    STR     R1, [R0, #GPIO_MODER]

; --- Cache base addresses ---
    LDR     R5, =GPIOB_BASE

; --- Load array addresses ---
    LDR     R6, =seg_patterns
    LDR     R7, =digit_select

; --- R8 = current count (0-9) ---
    MOV     R8, #0

; --- R9 = 1 second tick counter ---
    LDR     R9, =ONE_SECOND

main_loop

; --- Load segment pattern for current count ---
    LDRB    R10, [R6, R8]

; --- Load digit select (always digit 1) ---
    LDRH    R11, [R7]

; --- Combine and write to 7-seg ODR ---
    ORR     R10, R10, R11
    LDR     R1, =0x00000FFF
    LDR     R2, [R5, #GPIO_ODR] ; R5 = port B
```

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Target_1

task3.s*

```
main_loop

; --- Load segment pattern for current count ---
    LDRB    R10, [R6, R8]

; --- Load digit select (always digit 1) ---
    LDRH    R11, [R7]

; --- Combine and write to 7-seg ODR ---
    ORR     R10, R10, R11
    LDR     R1, =0x00000FFF
    LDR     R2, [R5, #GPIO_ODR] ; R5 = port B
    BIC     R2, R2, R1          ; clears lower 12 bits
    ORR     R2, R2, R10         ;(12 pins on 4-digit 7 segment display
    STR     R2, [R5, #GPIO_ODR]

; --- MUX delay (keep digit lit) ---
    LDR     R3, =MUX_DELAY
mux_delay_loop
    SUBS    R3, R3, #1
    BNE     mux_delay_loop

; --- Blank 7-seg ---
    LDR     R1, =0x00000FFF
    LDR     R2, [R5, #GPIO_ODR]
    BIC     R2, R2, R1          ; clear lower 12 bits
    STR     R2, [R5, #GPIO_ODR]

; --- Decrement 1 second counter ---
    SUBS    R9, R9, #1
    BNE     main_loop           ; not 1 second yet, keep multiplexing

; --- 1 second elapsed - advance count ---
    LDR     R9, =ONE_SECOND     ; reset tick counter
    ADD     R8, R8, #1          ; increment count
    CMP     R8, #10
    MOVGE   R8, #0              ; wrap back to 0 after 9

    B       main_loop

deadloop
    B       deadloop
```

Simulati

Part C

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Target_1

task3.s task1.c task2.s

```
    AREA mydata, DATA, READONLY

; Segment patterns for 0-9, active HIGH common cathode
; Bit order: dp g f e d c b a
seg_patterns    DCB     0x3F    ; 0 = a,b,c,d,e,f
                DCB     0x06    ; 1 = b,c
                DCB     0x5B    ; 2 = a,b,d,e,g
                DCB     0x4F    ; 3 = a,b,c,d,g
                DCB     0x66    ; 4 = b,c,f,g
                DCB     0x6D    ; 5 = a,c,d,f,g
                DCB     0x7D    ; 6 = a,c,d,e,f,g
                DCB     0x07    ; 7 = a,b,c
                DCB     0x7F    ; 8 = a,b,c,d,e,f,g
                DCB     0x6F    ; 9 = a,b,c,d,f,g

; Active LOW digit select - only using digit 1 (CA1)
; CA1=PB8 LOW, rest HIGH
digit_select    DCW     0x0E00

; Register definitions
RCC_AHB2ENR     EQU     0x4002104C      ; enable port clock register
GPIOAEN         EQU     (1<<0)          ; port A 0x00000001
GPIOBEN         EQU     (1<<1)          ; port B 0x00000002

GPIOA_BASE      EQU     0x48000000
GPIOB_BASE      EQU     0x48000400
GPIO_MODER      EQU     0x00            ; MODER offset
GPIO_ODR        EQU     0x14            ; ODR   offset

;
; adjust to 80000000 if running at 80MHz
ONE_SECOND      EQU     1000
MUX_DELAY       EQU     2000

    AREA mycode, CODE, READONLY
    EXPORT __main
```

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Target_1

task3.s | task1.c | task2.s

```
    AREA mycode, CODE, READONLY
    EXPORT __main
    ENTRY

__main

; --- Enable GPIOA and GPIOB clocks ---
    LDR     R0, =RCC_AHB2ENR
    LDR     R1, [R0]                ; 32 bit value in AHB2ENR
    ORR     R1, R1, #GPIOAEN        ; or with 0x00000001
    ORR     R1, R1, #GPIOBEN        ; or with 0x00000002
    STR     R1, [R0]                ; store in AHB2ENR

; --- Configure PA0-PA7 as outputs (LEDs) ---
    LDR     R0, =GPIOA_BASE
    LDR     R1, [R0, #GPIO_MODER]
    LDR     R2, =0x0000FFFF         ; set lower 16 bits as 1's
    BIC     R1, R1, R2              ; clears 16 lower bits                    5
    LDR     R2, =0x00005555         ; sets lower 8 pairs of 2 bits as 01 EX: 0101 (
    ORR     R1, R1, R2
    STR     R1, [R0, #GPIO_MODER]

; --- Configure PB0-PB11 as outputs (7-seg) ---
    LDR     R0, =GPIOB_BASE
    LDR     R1, [R0, #GPIO_MODER]
    LDR     R2, =0x00FFFFFF
    BIC     R1, R1, R2
    LDR     R2, =0x00555555
    ORR     R1, R1, R2
    STR     R1, [R0, #GPIO_MODER]

; --- Cache base addresses ---
    LDR     R4, =GPIOA_BASE
    LDR     R5, =GPIOB_BASE

; --- Load array addresses ---
    LDR     R6, =seg_patterns
    LDR     R7, =digit_select

; --- R8 = current count (0-9) ---
    MOV     R8, #0
```

ST-Link Debugge

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Target_1

task3.s task1.c task2.s

```
; --- R8 = current count (0-9) ---
    MOV     R8, #0

; --- R9 = 1 second tick counter ---
    LDR     R9, =ONE_SECOND

; ============================================================
; Main loop - multiplex 7-seg while counting
; ============================================================
main_loop

; --- Update LEDs with binary value of current count ---
    LDR     R1, [R4, #GPIO_ODR] ; R4 = port A
    BIC     R1, R1, #0xFF       ; clears lower 8 bits
    ORR     R1, R1, R8          ; R8 = count, maps directly to PA0-PA7
    STR     R1, [R4, #GPIO_ODR]

; --- Load segment pattern for current count ---
    LDRB    R10, [R6, R8]

; --- Load digit select (always digit 1) ---
    LDRH    R11, [R7]

; --- Combine and write to 7-seg ODR ---
    ORR     R10, R10, R11
    LDR     R1, =0x00000FFF
    LDR     R2, [R5, #GPIO_ODR] ; R5 = port B
    BIC     R2, R2, R1          ; clears lower 12 bits
    ORR     R2, R2, R10         ;(12 pins on 4-digit 7 segment display
    STR     R2, [R5, #GPIO_ODR]

; --- MUX delay (keep digit lit) ---
    LDR     R3, =MUX_DELAY
mux_delay_loop
    SUBS    R3, R3, #1
    BNE     mux_delay_loop

; --- Blank 7-seg ---
    LDR     R1, =0x00000FFF
    LDR     R2, [R5, #GPIO_ODR]
    BIC     R2, R2, R1          ; clear lower 12 bits
    STR     R2, [R5, #GPIO_ODR]
```

ST-Link Debugge

```
; --- Blank 7-seg ---
    LDR     R1, =0x00000FFF
    LDR     R2, [R5, #GPIO_ODR]
    BIC     R2, R2, R1          ; clear lower 12 bits
    STR     R2, [R5, #GPIO_ODR]

; --- Decrement 1 second counter ---
    SUBS    R9, R9, #1
    BNE     main_loop           ; not 1 second yet, keep multiplexing

; --- 1 second elapsed - advance count ---
    LDR     R9, =ONE_SECOND     ; reset tick counter
    ADD     R8, R8, #1          ; increment count
    CMP     R8, #10
    MOVGE   R8, #0              ; wrap back to 0 after 9

    B       main_loop

deadloop
    B       deadloop

    END
```

ST-Link Debugge